# Haonan Chen | From Robot Intelligence to Scientific Discovery: Building AI Agents that Think Like Scientists

## Research Vision

The next frontier of artificial intelligence lies not just in automation, but in enhancing human curiosity and creativity. My proposed research reimagines AI agents as *scientific collaborators*—systems that help scientists explore new ideas, design better experiments, and accelerate discovery in fields such as materials science, chemistry, and climate modeling. Rather than solving predefined tasks, these agents will explore strategically: proposing mechanisms, designing experiments, and refining their understanding over time. This represents a fundamental shift in AI's role—from executing decisions to participating in the generation of knowledge. By leveraging my expertise in robot learning and physical reasoning, this research aims to transition AI from automation to active, collaborative discovery.

Imagine a materials scientist trying to discover a new colloidal structure with targeted mechanical properties. Traditionally, this would require simulating thousands of variations and manually analyzing results. With the proposed system, an AI agent could model physical constraints, propose high-potential candidates, and refine its suggestions as new simulations or experiments are conducted—mirroring the reasoning process of an expert collaborator.

The research lays the foundation for AI agents that not only solve problems, but uncover the ones worth solving. I hypothesize that agents emulating scientific reasoning can accelerate discovery in domains like materials science and chemistry. To evaluate this, I will develop a modular framework and apply it to colloidal self-assembly and catalytic reaction design, working closely with scientific collaborators to ensure real-world impact and measurable improvements over current approaches.

## Interdisciplinary Pivot from Ph.D. Research

During my Ph.D., I developed learning-based robotic agents capable of generalizing physical interactions—such as tool use or object manipulation—across tasks, embodiments, and viewpoints. These agents learned to act in the world through data-driven, physically grounded models.

My proposed postdoctoral research represents a fundamental pivot: from developing agents focused on control to creating ones capable of scientific reasoning, insight generation, and collaborative discovery. No longer focused on control or imitation, these agents will engage in the reasoning loop of science: proposing candidate mechanisms, designing informative simulations or experiments, and adapting hypotheses in response to evidence.

This pivot introduces:

- A shift in goal: from performance-driven control to open-ended scientific exploration
- A shift in collaborators: from engineers to scientists
- A shift in data: from task trajectories to structured, noisy scientific observations

While my foundation in learning physical systems carries over, the intellectual tools, evaluation frameworks, and research culture are entirely new. This pivot is not just a shift in application, but a genuine disciplinary pivot—from embodied intelligence to scientific cognition—requiring new methods, collaborators, and a redefined vision of success. This

new direction not only broadens my methodological toolkit, but also positions me to impact how science is conducted.

## Proposed Methodology and Applications

To realize this pivot, I will build a set of modular AI agents that emulate core components of the scientific method. This research will develop modular AI agents that emulate the scientific process: forming hypotheses, testing predictions, and refining their understanding of complex systems. The architecture includes:

- **Learning Physical Intuition:** A model that captures how scientific systems behave—for example, how particles self-assemble or how chemical reactions proceed. These models will draw on my robotics work in learning intuitive physical representations and will be adapted to scientific domains.
- **Hypothesis Generator:** A module that proposes new candidate solutions—such as molecular structures or experimental conditions. I will explore generative models that learn to propose ideas informed by physical constraints and domain-specific priors.
- **Experiment Planner:** A decision-making module that selects the most informative simulations or experiments to perform next. Inspired by how human researchers plan studies, this system will optimize exploration to maximize insight under limited resources.

I will evaluate these agents in two high-impact domains:

- **Colloidal Self-Assembly:** Designing colloidal interaction potentials (such as anisotropic or patchy particles) to reliably self-assemble into desired crystal lattices. Success will be measured by the quality, stability, and novelty of assembled structures compared to baseline simulation-guided methods.
- **Catalytic Reaction Pathways:** Accelerating the identification of stable intermediates and reaction pathways for $CO_2$ reduction, particularly for C–C coupling steps on copper catalysts. The agent will propose candidate pathways that are then evaluated via DFT (Density Functional Theory) energy profiling or surrogate models trained on quantum chemical data.

These domains were chosen because they represent real-world challenges in scientific exploration where simulations are expensive, and the search space is vast. I will work with domain collaborators to tailor the systems to their workflows, ensuring the agents are not just theoretically sound but practically useful.

For instance, an AI collaborator could propose a counterintuitive combination of colloidal interactions that leads to a previously unknown quasicrystalline phase—one that evaded human intuition due to its unconventional symmetry. Such agents could uncover new scientific frontiers.

### Evaluation Timeline

To ensure this system is developed and evaluated effectively within the Fellowship, I have outlined a two-year plan below.

- **Months 1–6:** Co-design problems with scientific collaborators, adapt intuition models to new domains.

- **Months 6–12:** Prototype the hypothesis generator and experiment planner on simulation benchmarks.
- **Months 12–18:** Deploy agent in real-world workflows with domain experts.
- **Months 18–24:** Open-source toolkits and publish case studies demonstrating novel discoveries.

## New Skills and Tools to Acquire

To successfully execute this pivot, I will acquire new competencies, including:

- **Domain knowledge** in materials science and computational chemistry, including scientific design goals and key modeling assumptions.
- **Active learning and uncertainty estimation** to efficiently guide experiment selection and model improvement.
- **Interdisciplinary collaboration skills,** including co-designing research questions with scientists, translating between disciplines, and contributing meaningfully to lab-based workflows.
- **Scientific reasoning frameworks**, including hypothesis-driven design, falsifiability, and domain-specific evaluation metrics—concepts rarely emphasized in robotics.
- **Computational modeling techniques**, such as coarse-grained molecular dynamics for colloidal systems and DFT-based approaches for reaction energy profiling.

I intend to join a leading laboratory in a scientific domain (e.g., materials science or chemical engineering) to embed myself in day-to-day research. This will ensure that the tools I build are driven by real needs and integrated into real discovery pipelines.

## Potential Impact: Scientific and Societal, Short- and Long-Term

In the short term, this research delivers usable tools for scientific exploration. In the long term, it lays the foundation for a new generation of AI collaborators that can transform how scientific knowledge is generated across disciplines.

**Scientific Impact:** This research builds a new class of AI systems—those capable of engaging in scientific reasoning. These agents will not just optimize discovery pipelines; they will challenge how we think about automation, creativity, and knowledge generation itself.

**Societal Impact:** By enabling faster, more principled discovery in domains like energy, materials, and climate, this work contributes directly to solving pressing global challenges. All models and tools will be released open-source to support transparency and foster broader innovation.

## Alignment with the Schmidt Science Fellows Vision

This research embraces the core goals of the Schmidt Science Fellows program. It is a genuine disciplinary pivot—from robotics to scientific discovery—with meaningful intellectual risk and the potential for transformative impact. It brings AI into the heart of the scientific process, not to replace researchers, but to empower them.

Through this pivot, I aim to help define the next generation of AI—agents that learn not just how to act, but how to **think with us**, **reason with us**, and **discover with us**.